4.1

PAEDIATRICS

Bone and Joint Infections

CHAPTER 4.1	BONE AND JOINT INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
Septic arthritis			
Neonates			
Group B streptococcus MSSA Gram-negative enteric organisms *Candida albicans* (on TPN)	Cefotaxime 50 mg/kg/dose IV q12H (<7 days of life) q6H (>7 days of life) + Cloxacillin 50 mg/kg/dose IV q12H (<7 days of life) q6H (>7 days of life)		• Final choice of antibiotics depends on C&S of pus obtained from joint aspiration or at surgery.
Older Children			

CHAPTER 4.1	BONE AND JOINT INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
MSSA Group A *Streptococcus* *S pneumoniae* *Haemophilus spp* (non-typable) *Kingella kingae*	**<5 years old:** Co-amoxiclav 25 mg/kg/dose (Amoxicillin component) IV q8H	**<5 years old:** Cefazolin 50mg/kg/dose IV q8H **OR** Cefuroxime 30 mg/kg/dose IV q8H	• IV to PO switch is permissible upon good clinical and laboratory response. • Total duration: 3-4 weeks.
MSSA	**>5 years old:** Cloxacillin 50 mg/kg/dose IV q6H	**>5 years old:** Cefazolin 50mg/kg/dose IV q8H	
MRSA	Vancomycin 15 mg/kg/dose IV q8H + Rifampicin 7.5-10 mg/kg/dose (max 600 mg/day) PO q12H	Clindamycin 10mg/kg/dose IV q8H** **OR** Linezolid 10 mg/kg/dose IV (max 600 mg) q8H	• Vancomycin penetrates poorly into synovial space. • **For community-acquired methicillin-resistant *S. aureus* (CA-MRSA) only.
Acute osteomyelitis			

CHAPTER 4.1	BONE AND JOINT INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
Neonates			
S. aureus Group B Streptococci Coliforms *Pseudomonas aeruginosa* *Candida spp.*	Cloxacillin 50 mg/kg/dose IV q12H (<7 days of life) q6H (>7 days of life) + Cefotaxime 50 mg/kg/dose IV q12H (<7 days of life) q6H (>7 days of life)	**For MRSA:** Linezolid 10 mg/kg/dose IV (max 600 mg) q8H	• Final choice of antibiotics depends on C&S of pus/tissue.
Older Children			
S. aureus Group A Streptococcus *Haemophilus spp.* *S. pneumoniae* Anaerobes	**≤5 years old:** Co-amoxiclav 25 mg/kg/dose IV (Amoxicillin component) q8H		• Final choice of antibiotics depends on C&S of pus/ tissue • IV to oral switch permissible upon good clinical response and at least 2 weeks of IV antibiotic.
	>5 years old:		• **Duration:**

CHAPTER 4.1	BONE AND JOINT INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
	Cloxacillin 50 mg/kg/dose IV q6H	Cefazolin 50 mg/kg/dose IV q8H	• 4 weeks if uncomplicated (single bone, rapid response to treatment, rapid resolution of bacteraemia, no sequelae i.e., endocarditis, thrombosis, physeal injury). • 6 weeks or longer if complicated (multifocal, additional soft tissues infection, unusual pathogens, slow clinical response, multiple surgical interventions, sequelae i.e., endocarditis, thrombosis, pathologic fracture, concern of physeal injury).
Chronic osteomyelitis			
S. aureus (MSSA)	Cloxacillin 25 mg/kg/dose PO q6H	Cefalexin 100 mg/kg/day PO q8H	• Surgical debridement is recommended. • Final choice of antibiotics depends on C&S of pus/tissue. • Consider adding Rifampicin for prosthetic joint infection.